A SHORT NOTE ON NILPOTENT AND IDEMPOTENT REAL MATRICES

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Abstract. We give a self-contained account of the kernel chain of a nilpotent real matrix and deduce the classical bound on its index of nilpotency. We also collect a few facts about idempotents and commutators that are used in teaching a second course in linear algebra.

1. Introduction

Nilpotent and idempotent matrices are the two basic building blocks in the Jordan and Fitting decompositions of a linear operator. Many textbooks prove that a nilpotent $n \times n$ matrix A satisfies $A^n = 0$ only after developing the Jordan normal form or the Cayley–Hamilton theorem. The purpose of this note is to record a direct argument based on the chain of kernels $\ker A \subseteq \ker A^2 \subseteq \cdots$, which uses nothing beyond the dimension of a subspace (Theorem 2.5). Along the way we record some elementary facts on sums of commuting nilpotent matrices, on idempotents, and, in Section 3, an application of a classical lemma of Jacobson to commutators.

Throughout, $n \geq 1$ is an integer, $M_n(\mathbb{R})$ is the algebra of real $n \times n$ matrices, I is the identity matrix, and matrices act on column vectors in $\mathbb{R}^n$. We write $\ker A$, $\operatorname{im} A$, $\operatorname{tr} A$ and $\operatorname{rank} A$ for the kernel, image, trace and rank of A, and $A^0 = I$.

2. The kernel chain and the index of nilpotency

Definition 2.1. A matrix $A \in M_n(\mathbb{R})$ is *nilpotent* if $A^k = 0$ for some integer $k \geq 1$; the least such k is the *index* of A and is denoted $\nu(A)$. A matrix $P \in M_n(\mathbb{R})$ is *idempotent* if $P^2 = P$. For $A \in M_n(\mathbb{R})$ and $k \geq 0$ we put $d_k = d_k(A) = \dim \ker A^k$.

Lemma 2.2. *Let $A \in M_n(\mathbb{R})$. Then $\ker A^k \subseteq \ker A^{k+1}$ for every $k \geq 0$. In particular $0 = d_0 \leq d_1 \leq d_2 \leq \cdots \leq n$.*

Proof. If $A^k x = 0$ then $A^{k+1}x = A(A^k x) = 0$. The second statement follows since $\ker A^0 = \ker I = 0$ and every $\ker A^k$ is a subspace of $\mathbb{R}^n$. □

Lemma 2.3. *Let $A \in M_n(\mathbb{R})$ and $k \geq 0$. If $\ker A^k = \ker A^{k+1}$, then $\ker A^{k+j} = \ker A^k$ for every $j \geq 0$.*

Proof. By induction on j it suffices to show that $\ker A^{k+1} = \ker A^{k+2}$. The inclusion $\subseteq$ is Lemma 2.2. Conversely, let $x \in \ker A^{k+2}$. Then $A^{k+1}(Ax) =$

0, so $Ax \in \ker A^{k+1} = \ker A^k$, that is, $A^k(Ax) = 0$. Hence $A^k x = 0$, so $x \in \ker A^k = \ker A^{k+1}$, as required. □

Lemma 2.4. *Let $A \in M_n(\mathbb{R})$ be nilpotent. Then $d_k < d_{k+1}$ for every $0 \leq k \leq \nu(A)$.*

Proof. Write $\nu = \nu(A)$ and suppose that $d_k = d_{k+1}$ for some k in the stated range. Since $\ker A^k \subseteq \ker A^{k+1}$ by Lemma 2.2, equal dimensions give $\ker A^k = \ker A^{k+1}$, and Lemma 2.3 yields $\ker A^k = \ker A^{k+j}$ for all $j \geq 0$. Taking j with $k + j \geq \nu$ we get $\ker A^k = \ker A^{k+j} = \mathbb{R}^n$, i.e. $A^k = 0$. If $k = 0$ this says $I = 0$, which is impossible as $n \geq 1$; if $k \geq 1$ it contradicts the minimality of ν. □

Theorem 2.5. *Let $A \in M_n(\mathbb{R})$ be nilpotent. Then $\nu(A) \leq n$. In particular $A^n = 0$.*

Proof. Write $\nu = \nu(A)$. By Lemma 2.4 applied with $k = 0, 1, \ldots, \nu - 1$ we have $0 = d_0 < d_1 < \cdots < d_\nu$, so $d_\nu \geq \nu$. Since $d_\nu \leq n$ by Lemma 2.2, we obtain $\nu \leq n$. Finally $A^n = A^{n-\nu} A^\nu = 0$. □

Proposition 2.6. *Let $N \in M_n(\mathbb{R})$ be nilpotent with $\nu = \nu(N)$, and let $x \in \mathbb{R}^n$ satisfy $N^\nu x \neq 0$. Then the vectors $x, Nx, \ldots, N^{\nu-1}x$ are linearly independent. Consequently the subspace they span is N-invariant of dimension ν.*

Proof. Note first that $N^{\nu-1}x \neq 0$, since $N^\nu x = N(N^{\nu-1}x)$. Suppose $\sum_{i=0}^{\nu-1} c_i N^i x = 0$ with not all c_i zero, and let j be the least index with $c_j \neq 0$. Applying $N^{\nu-1-j}$ and using $N^m = 0$ for $m \geq \nu$ leaves $c_j N^{\nu-1} x = 0$, a contradiction. The span is N-invariant because $N(N^i x) = N^{i+1}x$ for $i < \nu - 1$ and $N(N^{\nu-1}x) = N^\nu x = 0$. □

Lemma 2.7. *Let $A, B \in M_n(\mathbb{R})$ be nilpotent with $AB = BA$. Then $A + B$ is nilpotent and $\nu(A + B) \leq \max\{\nu(A), \nu(B)\}$.*

Proof. Let $m = \max\{\nu(A), \nu(B)\}$. Since A and B commute, the binomial theorem gives $(A + B)^m = \sum_{i=0}^m \binom{m}{i} A^i B^{m-i}$. In each term either $i \geq \nu(A)$ or $m - i \geq \nu(B)$, so every term vanishes and $(A + B)^m = 0$. □

Proposition 2.8. *Let $P \in M_n(\mathbb{R})$ be idempotent and symmetric, $P^\mathsf{T} = P$. Then $\operatorname{tr} P = \operatorname{rank} P$.*

Proof. Every $x \in \mathbb{R}^n$ can be written $x = Px + (x - Px)$ with $Px \in \operatorname{im} P$ and $P(x - Px) = Px - P^2 x = 0$; and if $x \in \operatorname{im} P \cap \ker P$, say $x = Py$, then $x = Py = P^2 y = Px = 0$. Hence $\mathbb{R}^n = \operatorname{im} P \oplus \ker P$. Moreover P acts as the identity on $\operatorname{im} P$, since $P(Py) = Py$. Choosing a basis of $\mathbb{R}^n$ formed by a basis of $\operatorname{im} P$ followed by a basis of $\ker P$, the matrix P becomes similar to $\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$ with $r = \operatorname{rank} P$. As the trace is invariant under similarity, $\operatorname{tr} P = r$. □

Lemma 2.9. *Let $P \in M_n(\mathbb{R})$ be idempotent and let $t \in \mathbb{R}$ with $t \neq -1$. Then $I + tP$ is invertible and*

$$(I + tP)^{-1} = I - \frac{t}{1+t}\,P.$$

Proof. Using $P^2 = P$, $(I + tP)\big(I - \frac{t}{1+t}P\big) = I + \big(t - \frac{t}{1+t} - \frac{t^2}{1+t}\big)P = I$, and the product in the other order is computed in the same way. □

3. An application to commutators

For $A, B \in M_n(\mathbb{R})$ we write $[A, B] = AB - BA$. The following classical result is due to Jacobson; we state it in the finite-dimensional form over a field of characteristic zero, see [1].

Theorem 3.1 (Jacobson [1])**.** *Let $A, B \in M_n(\mathbb{R})$ and put $C = [A, B]$. If $AC = CA$, then C is nilpotent; in fact $C^2 = 0$.*

Corollary 3.2. *Let $A, B \in M_n(\mathbb{R})$ and suppose that $[A, B]$ commutes with $A + B$. Then $[A, B]$ is nilpotent; in particular $[A, B]^n = 0$.*

Proof. Put $X = A + B$. Then $[X, B] = [A, B] + [B, B] = [A, B]$, so the hypothesis says that $[X, B]$ commutes with X. By Theorem 3.1 applied to the pair (X, B), the matrix $[X, B] = [A, B]$ is nilpotent, and Theorem 2.5 gives $[A, B]^n = 0$. □

Remark 3.3. Corollary 3.2 is symmetric in A and B, as it should be: $[A, B]$ commutes with $A + B$ if and only if $[B, A] = -[A, B]$ does.

References

[1] N. Jacobson, *Rational methods in the theory of Lie algebras*, Ann. of Math. (2) **36** (1935), 875–881.

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